

Barcoding Score Test

An Interpretable Sliding-Window Detection Concept

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Barcoding Score Test

Objective Develop an interpretable, classical image-analysis detector that assigns a continuous barcoding score to each horizontal position of a processed OCT B-scan.

General workflow

E2E volume → Preprocess & Denoise → Feature Engineering → Barcoding Score

The initial detector evaluates two forms of evidence:

- **Structural evidence:** vertical organization and repeated spatial texture.
- **Hypertransmission evidence:** brightness and persistence through depth.

This first run is intended as a proof of concept. No final clinical threshold, feature weights, or parameter values have yet been tuned.

Preprocessing Pipeline

For each selected E2E volume:

- 1 Load, then Select the center B-scan using

$$i_{\text{center}} = \left\lfloor \frac{N_{\text{B-scans}}}{2} \right\rfloor.$$

- 2 Retrieve the BM boundary and interpolate missing boundary coordinates.
- 3 Flatten the image so BM lies on one horizontal reference row.
- 4 Crop a fixed region extending 150 pixels below BM.
- 5 Apply whole-ROI z-score normalization.
- 6 Apply mild anisotropic Gaussian denoising.

The resulting detector input has approximate dimensions $150(h) \times 512(w)$

Whole-ROI Z-Score Normalization

After flattening, the entire sub-BM region is normalized using

$$Z(z, x) = \frac{I_{\text{ROI}}(z, x) - \mu_{\text{ROI}}}{\sigma_{\text{ROI}} + \varepsilon},$$

where: $I_{\text{ROI}}(z, x)$ is the intensity and $\varepsilon = 10^{-8}$ prevents division by zero.

For the first test scan:

$$\mu_{\text{ROI}} \approx 60.98, \quad \sigma_{\text{ROI}} \approx 44.73.$$

Normalization is performed once over the whole ROI; the sliding-window features do not renormalize each window.

Gaussian (Pseudo-ImageJ) Denoising

A two-dimensional anisotropic Gaussian filter is applied after normalization:

$$I_{\sigma_z, \sigma_x} = G_{\sigma_z, \sigma_x} * Z,$$

where:

- G_{σ_z, σ_x} is a Gaussian smoothing kernel,
- $*$ denotes convolution,
- σ_z controls smoothing along depth, 1.0 first used
- σ_x controls smoothing horizontally, 0.5 first used

Reason for asymmetric smoothing

- More smoothing is applied through depth to reduce OCT speckle.
- Less horizontal smoothing is applied to avoid erasing narrow vertical stripe boundaries.
- The observed denoising effect was intentionally subtle rather than strongly blurring the scan.

```
preprocess_bscan(volume, index, config)
```

Sliding-Window Configuration

The detector uses overlapping vertical windows that span the complete 150-pixel depth. For a window centered at horizontal position x ,

$$W_x = I[0 : 149, x - r : x + r],$$

where the initial window radius is $r=10$. Therefore,

$$\text{window width} = 2r + 1 = 21 \text{ pixels.}$$

The initial stride is a single pixel, so the windows are:

$$(0, \dots, 20), \quad (1, \dots, 21), \quad (2, \dots, 22), \quad \dots$$

Each window produces one feature vector assigned to its center position. Reflection padding is used at the horizontal image boundaries. The first and last 10 score positions are later excluded from detection to reduce edge artifacts.

Structural Feature 1: Verticality

Verticality is calculated using a **structure tensor**, which summarizes the direction of local image-intensity changes.

Let

$$I_x = \frac{\partial I}{\partial x}, \quad I_z = \frac{\partial I}{\partial z},$$

where:

- I_x measures horizontal brightness change,
- I_z measures brightness change through depth.

The local structure tensor is

$$J = \begin{bmatrix} \langle I_x^2 \rangle & \langle I_x I_z \rangle \\ \langle I_x I_z \rangle & \langle I_z^2 \rangle \end{bmatrix}.$$

Its eigenvalues satisfy $\lambda_1 \geq \lambda_2 \geq 0$. We calculate an anisotropy score as

$$A = \frac{\lambda_1 - \lambda_2}{\lambda_1 + \lambda_2 + \varepsilon}.$$

Verticality: Interpretation

The anisotropy score describes how strongly the local texture prefers one direction, with greater values indicating strong orientation. A vertical line usually produces strong horizontal intensity changes. Therefore, the verticality map combines:

- the strength of directional organization;
- agreement with the orientation expected from a vertical structure.

The window-level verticality score is the mean verticality across all pixels in the local 150-by-21 window.

Observed first-run behavior

- Verticality rose around visibly prominent vertical transmission columns.
- It was one of the strongest and most spatially interpretable features.
- Strong edge responses were also observed, motivating edge exclusion.

Structural Feature 2: Depth Continuity

Depth continuity measures whether the horizontal bright–dark pattern remains aligned as the image is traversed downward.

For each pair of neighboring depth rows, the detector calculates their horizontal correlation:

$$C_z = \text{corr}(I(z, :), I(z + 1, :)).$$

The window continuity score is the mean of the valid row-pair correlations:

$$C(x) = \frac{1}{N_{\text{valid}}} \sum_z C_z.$$

Here:

- $I(z, :)$ is the horizontal intensity pattern at depth row z ,
- $I(z + 1, :)$ is the pattern one row deeper,

The initial depth lag is a single row.

Depth Continuity: Interpretation

A barcode-like region contains vertical structures, so bright and dark horizontal positions should remain similar across successive depth rows.

Interpretation

- High continuity: the same column pattern persists through depth.
- Low continuity: local brightness patterns change substantially between depth rows.
- Negative continuity: horizontal patterns tend to reverse.

Observed first-run behavior

- Continuity aligned well with visually prominent vertical regions.
- It was substantially more stable than the initially tested periodicity feature.
- It was strongly correlated with verticality: ($r=0.917$)
- It was also strongly correlated with heterogeneity: ($r=0.920$)

These correlations motivated the grouped scoring formulation presented later.

Structural Feature 3: Spatial Heterogeneity

Spatial heterogeneity measures how much the local horizontal column intensity varies inside a window. First, each column is summarized across depth:

$$c(u) = \frac{1}{D} \sum_{z=0}^{D-1} I(z, u),$$

where:

- u is a horizontal column inside the local window,
- $D = 150$ is the ROI depth,
- $c(u)$ is the average intensity of that column.

The initial heterogeneity measure utilizes the standard deviation across the 21 column means.

Spatial Heterogeneity: Interpretation

Interpretation

- Low heterogeneity: neighboring columns have similar average intensity.
- High heterogeneity: the window contains alternating brighter and darker columns or abrupt spatial changes.

This may help distinguish:

- broad, relatively smooth hypertransmission;
- repeated or irregular vertical stripe structures.

Observed first-run behavior

Heterogeneity correlated highly with verticality (0.877) and continuity (0.920) so these three features are treated as related measurements of **structural organization**, not as fully independent evidence.

Hypertransmission Feature 1: Depth Persistence

Depth persistence measures how far candidate bright and vertically organized signal extends through the 150-pixel ROI.

A candidate pixel is defined (loosely) using:

- an intensity threshold at the 75th percentile of the processed scan;
- a minimum verticality requirement of 0.20.

For each column u , define a binary candidate signal whose longest-run persistence is

$$P(u) = \frac{L_{\max}(u)}{D},$$

where:

- $L_{\max}(u)$ is the longest uninterrupted candidate run,
- $D = 150$ is the full ROI depth.

The sliding-window persistence score is the mean across the 21 columns.

Depth Persistence: Interpretation

Interpretation

- $P(u) \approx 0$: little bright, vertical signal persists through depth.
- Larger $P(u)$: a longer continuous candidate signal is present.
- $P(u) \approx 1$: candidate signal extends through nearly the entire ROI depth.

Persistence is intended to distinguish:

- isolated bright pixels or shallow structures;
- sustained hypertransmission columns.

Observed first-run behavior

- Persistence increased near the strongest visible transmission regions.
- It was strongly related to amplitude:

$$r_{\text{persistence,amplitude}} = 0.823.$$

For this reason, persistence and amplitude form the **hypertransmission feature group**.

Hypertransmission Feature 2: Amplitude

Amplitude measures the average normalized intensity within each 150-by-21 window:

$$M(x) = \frac{1}{|W_x|} \sum_{(z,u) \in W_x} I(z, u).$$

Here:

- W_x is the window centered at horizontal position x ,
- $|W_x|$ is the number of pixels in the window,
- $I(z, u)$ is the normalized and denoised intensity.

Interpretation

- Higher amplitude indicates a brighter local sub-BM region.
- This is consistent with stronger hypertransmission.
- Amplitude alone is not sufficient for barcoding because EA may also appear broadly bright.

Therefore, amplitude is combined with structural features rather than used as a standalone classifier.

Robust Feature Standardization

The raw feature signals have different numerical scales. Before weighting, each feature is robustly standardized across the scan:

$$\tilde{F}_j(x) = \frac{F_j(x) - \text{median}(F_j)}{\text{IQR}(F_j) + \varepsilon}.$$

Here:

- $F_j(x)$ is feature j at position x ,
- $\text{median}(F_j)$ is the median across the scan,
- $\text{IQR}(F_j)$ is the 75th percentile minus the 25th percentile,
- $\varepsilon = 10^{-8}$ prevents division by zero.

The standardized values are clipped to $[-5,5]$ to limit the influence of extreme outliers.

This standardization makes the component magnitudes comparable before constructing the weighted score.

Individual Versus Grouped Scoring

Two scoring modes were implemented.

Individual scoring

$$S_{\text{individual}}(x) = \sum_{j=1}^5 w_j \tilde{F}_j(x).$$

Each of the five features receives its own direct weight.

Grouped scoring

Highly correlated features are first combined into two interpretable groups:

$$\begin{aligned} S_{\text{structural}}(x) &= w_V \tilde{V}(x) + w_C \tilde{C}(x) + w_H \tilde{H}(x), \\ S_{\text{hyperTD}}(x) &= w_P \tilde{P}(x) + w_A \tilde{A}(x). \end{aligned}$$

The final score is

$$S(x) = w_S S_{\text{structural}}(x) + w_T S_{\text{hyperTD}}(x).$$

The grouped mode was selected as the initial default.

Initial Individual Feature Weights

For the direct five-feature score, the initial proof-of-concept weights are:

Feature	Weight
Verticality	0.25
Persistence	0.25
Continuity	0.15
Amplitude	0.20
Heterogeneity	0.15

These weights sum to one.

They are not learned or clinically validated. They were chosen to:

- give substantial weight to vertical geometry;
- retain strong hypertransmission evidence;
- reduce, but not eliminate, redundant structural contributions.

Initial Grouped Weights

Within the structural group

$$S_{\text{structural}} = 0.45\tilde{V} + 0.275\tilde{C} + 0.275\tilde{H}.$$

Within the hypertransmission group

$$S_{\text{hyperTD}} = 0.55\tilde{P} + 0.45\tilde{A}.$$

Final group combination

$$S = 0.55S_{\text{structural}} + 0.45S_{\text{hyperTD}}.$$

This design reflects the first-scan correlation results:

- verticality, continuity, and heterogeneity formed one highly correlated structural cluster;
- persistence and amplitude formed a second hypertransmission cluster.

Grouping reduces the risk of counting closely related structural evidence three separate times.

Initial Threshold and Spatial Cleanup

A position is initially marked as barcode-positive when

$$S(x) > \tau.$$

The initial proof-of-concept threshold is $\tau = 0.0$. This threshold is provisional and has not yet been selected using manual annotations.

The raw one-dimensional mask is then cleaned using a minimum positive run of 8 positions and maximum gap of 4 positions, along with clipping the first 10 and last 10 positions to account for scanner error

The cleanup order is:

fill short gaps $\rightarrow$ remove short positive runs.

The final cleaned runs are converted into contiguous horizontal detection intervals.

First-Run Observations

The first scan showed encouraging qualitative agreement among multiple features.

Around the most visually prominent hypertransmission region, all engineered features increased, though how much each increases in EA over barcoding was not investigated.

The strongest observed feature correlations were:

$$r_{\text{continuity,heterogeneity}} = 0.920,$$

$$r_{\text{verticality,continuity}} = 0.917,$$

$$r_{\text{verticality,heterogeneity}} = 0.877,$$

$$r_{\text{persistence,amplitude}} = 0.823.$$

These results support the grouped score design, but they are based on only one scan and should not yet be treated as general conclusions.

① Expand beyond binary barcoding detection

- Characterize Normal, Early Atrophy, Barcoding, and possible vessel-related structures.
- Determine which mathematical features distinguish broad EA from narrow vertical barcode patterns.

② Run the pipeline across all available scans

- Process all B-scans within each E2E volume.
- Compare stability across neighboring scans from the same eye.
- Compare center scans and full volumes across subjects.

③ Validate against manual annotations

- Convert manually labeled intervals into column-level reference labels.
- Evaluate sensitivity, specificity, precision, F1 score, and interval overlap.

Future Weight and Threshold Tuning

After all available scans are processed, the dataset may contain at least 92×96 scan-level examples, with some volumes containing more than 96 B-scans.

A logistic regression model can then estimate feature weights once more scans have been processed and reliability is assessed.

Training and evaluation must be split by subject or volume rather than by individual image column to avoid leakage between highly related observations.

Thresholds can then be selected using validation criteria such as:

- F1 score;
- Youden's index;
- a clinically specified sensitivity target.

Immediate Next Steps

- ① Run the current untuned configuration on additional fast and slow progressor volumes.
- ② Inspect whether the five feature signals remain interpretable across different scans.
- ③ Compare grouped and individual scores.
- ④ Quantify within-volume consistency across neighboring B-scans.
- ⑤ Compare automatic intervals with manual Normal, EA, and Barcoding labels.
- ⑥ Perform sensitivity analysis for:
 - window width and stride;
 - Gaussian smoothing;
 - persistence thresholds;
 - feature weights;
 - score threshold;
 - spatial cleanup parameters.

The current results are a proof of concept, not a finalized detector.